

MISP Workbench

MISP Integration Workshop

What it is & why it exists

Architecture

Ingest, correlate, search, export

Enrichment, hunts & alerting

AI / MCP, reactors & notebooks

Hands-on

What is MISP Workbench?

A modern, MISP-compatible threat intelligence platform.

- Runs **standalone** — no full MISP instance required.
- Ingest feeds, **correlate** indicators, search, and export.
- Stays **compatible** with the wider MISP ecosystem.

Built for the analyst's investigation workflow, not just data entry.



2 / 15

Architecture

A few focused services working together.

- **FastAPI** backend exposes the API and UI.
- **OpenSearch** indexes attributes for fast search & correlation.
- **Celery** workers process feeds, hunts, and events asynchronously.
- **S3-compatible** or local storage for attachments.

Index-first design — everything is searchable the moment it lands.

Feed management

Bring intelligence in from anywhere.

- Ingest **MISP, CSV, JSON, and freetext** feeds.
- Run them **on a schedule** or trigger manually.
- New data is indexed and ready to correlate immediately.

One place to pull together every source you track.

Correlation & analysis

Find the overlaps that matter.

- **Batch** and **incremental** correlation scans over indexed attributes.
- Surface relationships and shared indicators across the dataset.
- Correlations update as new feed data arrives.

Correlation is where isolated indicators become a picture.

Search & exploration

Ask precise questions, get fast answers.

- **Lucene** query syntax against the OpenSearch index.
- Fast indicator lookups across the whole dataset.
- Pivot from a hit into related attributes and events.

The dataset is only as useful as your ability to query it.

Get intelligence back out in the right shape.

- Export in **JSON, CSV, MISP, or STIX 2.1**.
- Feed downstream tools, partners, and detection pipelines.
- Same data, multiple standards-compliant formats.

Interoperability keeps Workbench a good citizen of the ecosystem.

Enrichment

Add context automatically.

- IOC enrichment powered by **misp-modules**.
- Augment indicators with WHOIS, geolocation, reputation, and more.
- Turn a bare indicator into an actionable one.

Enrichment answers the "so what?" behind an indicator.

Hunts & alerting

Saved searches that keep working for you.

- **Hunts** are saved searches that run **periodically**.
- A match **triggers an alert** — no manual re-running.
- Notifications are processed by **Celery** workers.

Let the platform watch the feeds while you do other work.

Query threat intel from your AI assistant.

- Built-in **MCP Server** (Model Context Protocol).
- Ask **Claude, Cursor, and others** to query the dataset directly.
- Natural-language access to search, correlation, and context.

Bring the intelligence to the tools analysts already use.

Reactors & notebooks

Automate and explore with Python.

- **Reactor scripts** — Python that responds to platform events in a sandbox.
- **Interactive notebooks** for ad-hoc analyst exploration.
- Analytical tools pre-imported and ready to go.

Scripted automation for the routine, notebooks for the novel.

Hands-on

Hands-on: ingest & explore

Load a feed and query it end-to-end.

1. **Add a feed** and run it — watch attributes get indexed.
2. **Search** with a Lucene query and pivot on a hit.
3. **Correlate** to surface overlaps with existing data.

Full steps in the tutorial.

Hands-on: hunt & export

Turn a search into a standing alert, then share it.

1. **Save a search as a hunt** and let it run on a schedule.
2. **Trigger** a match and watch the alert fire.
3. **Export** the results as STIX 2.1 for a downstream tool.

Full steps in the tutorial.

Questions?